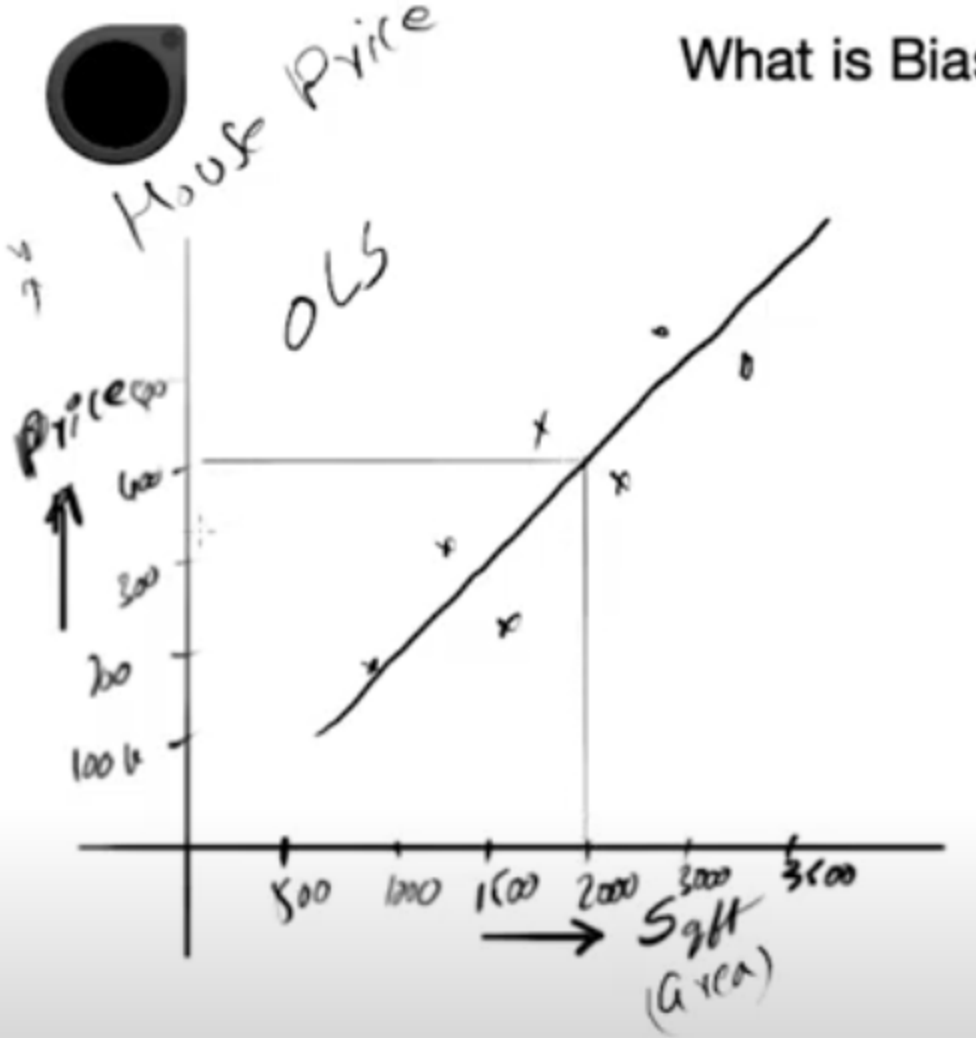
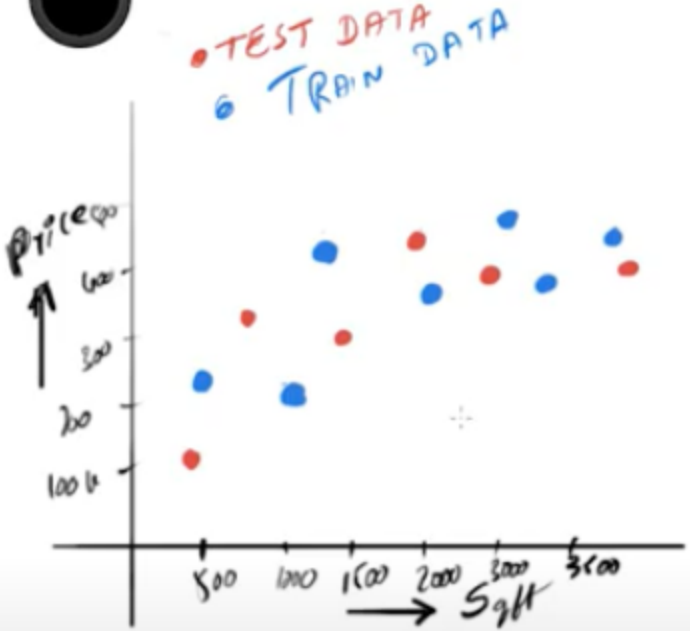
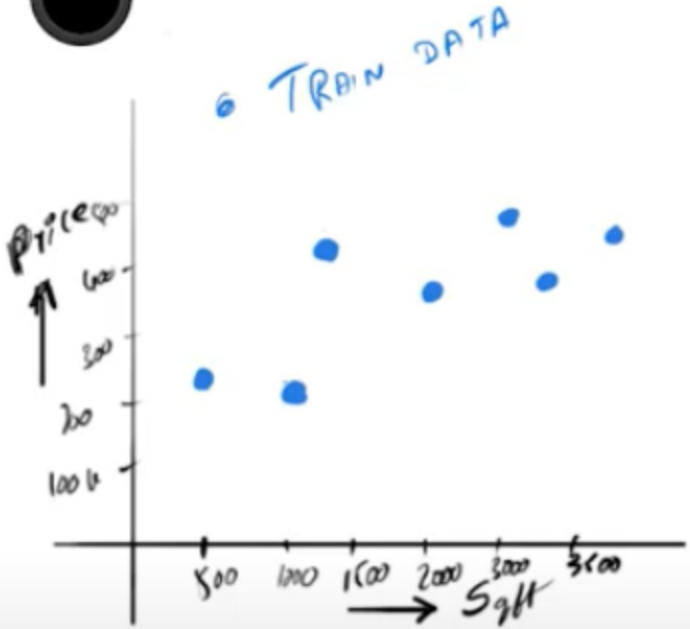
What is Bias-Variance Tradeoff?



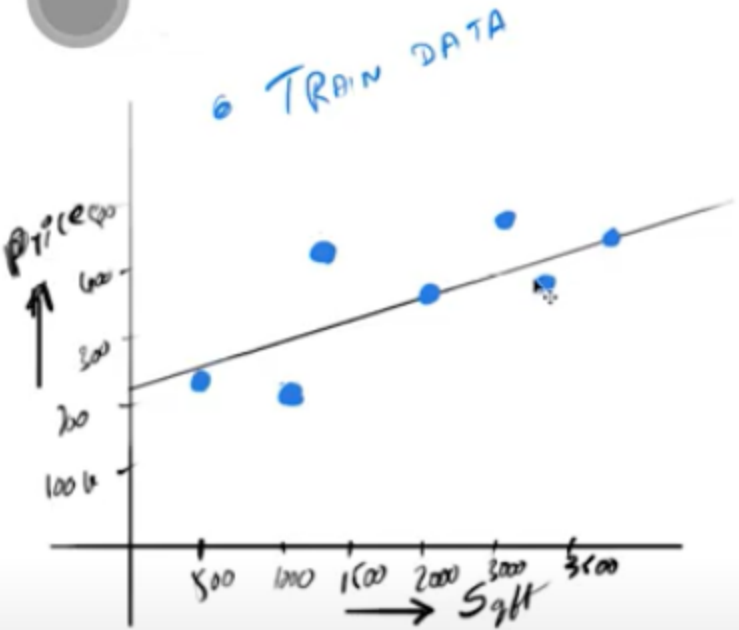
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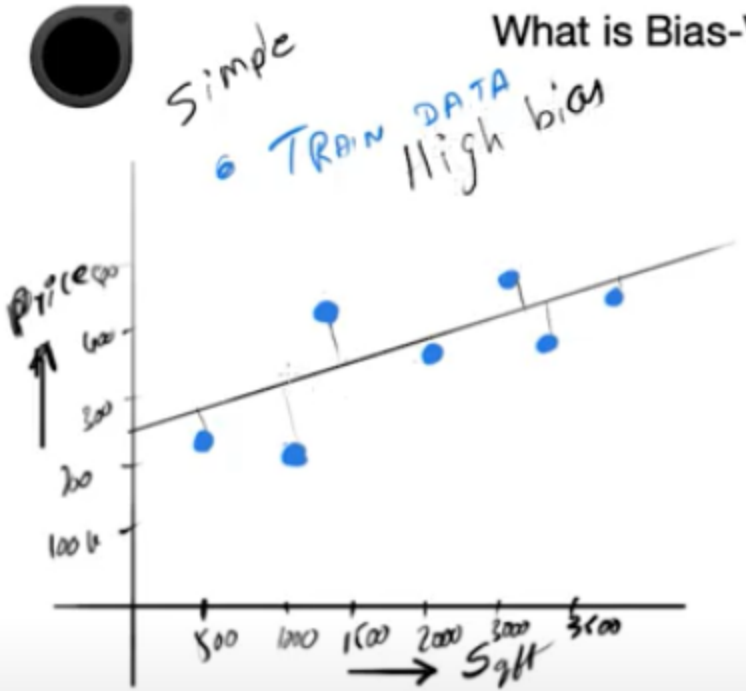
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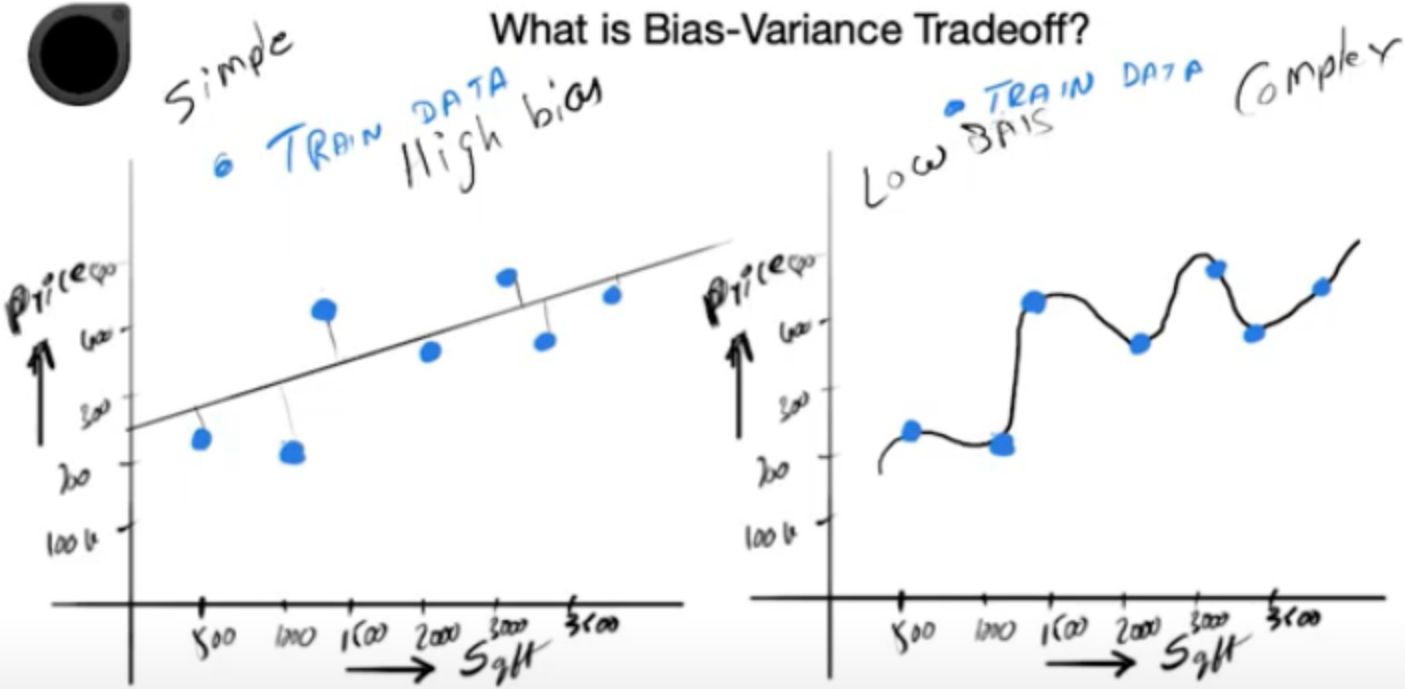
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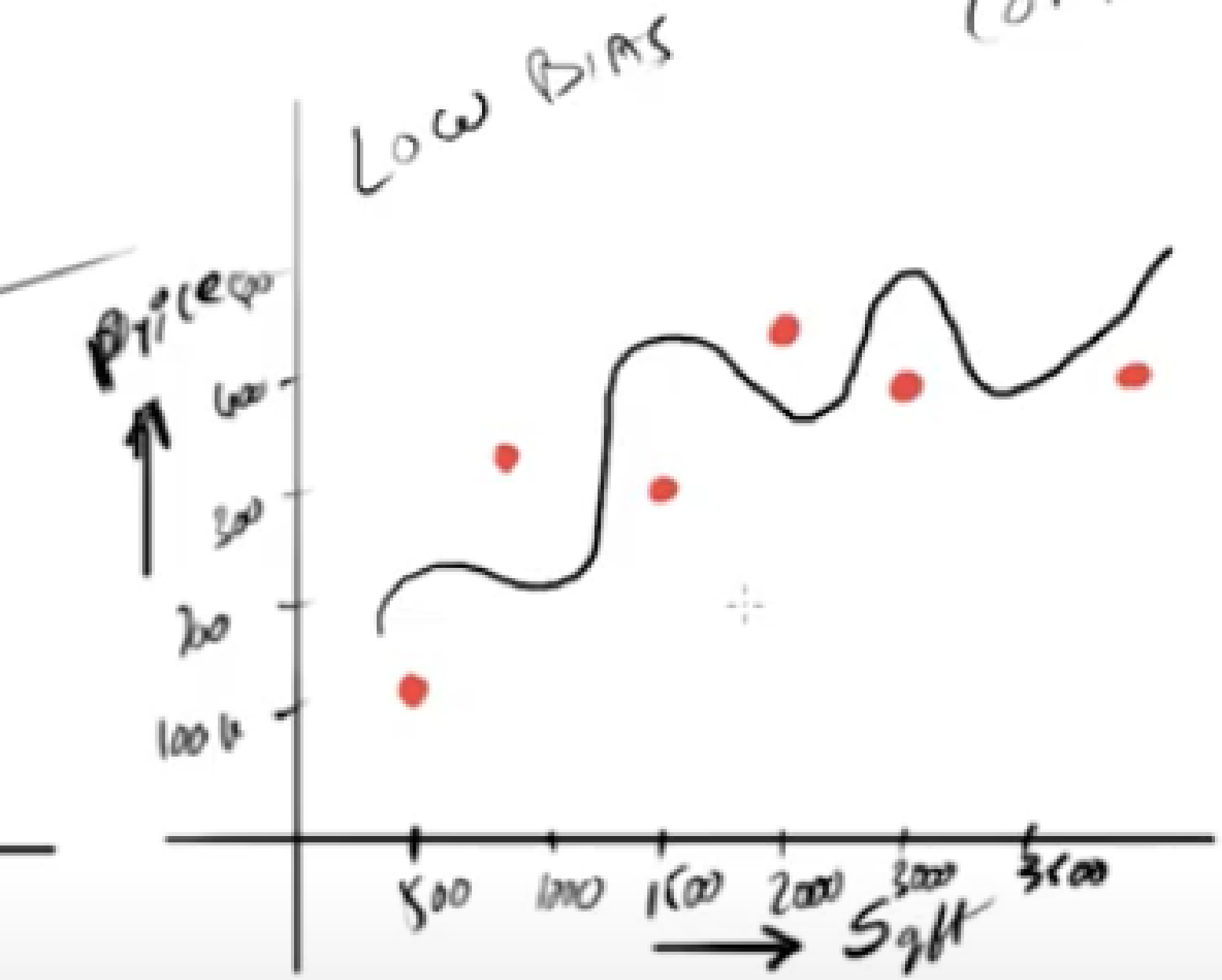


What is Bias-Variance Tradeoff?

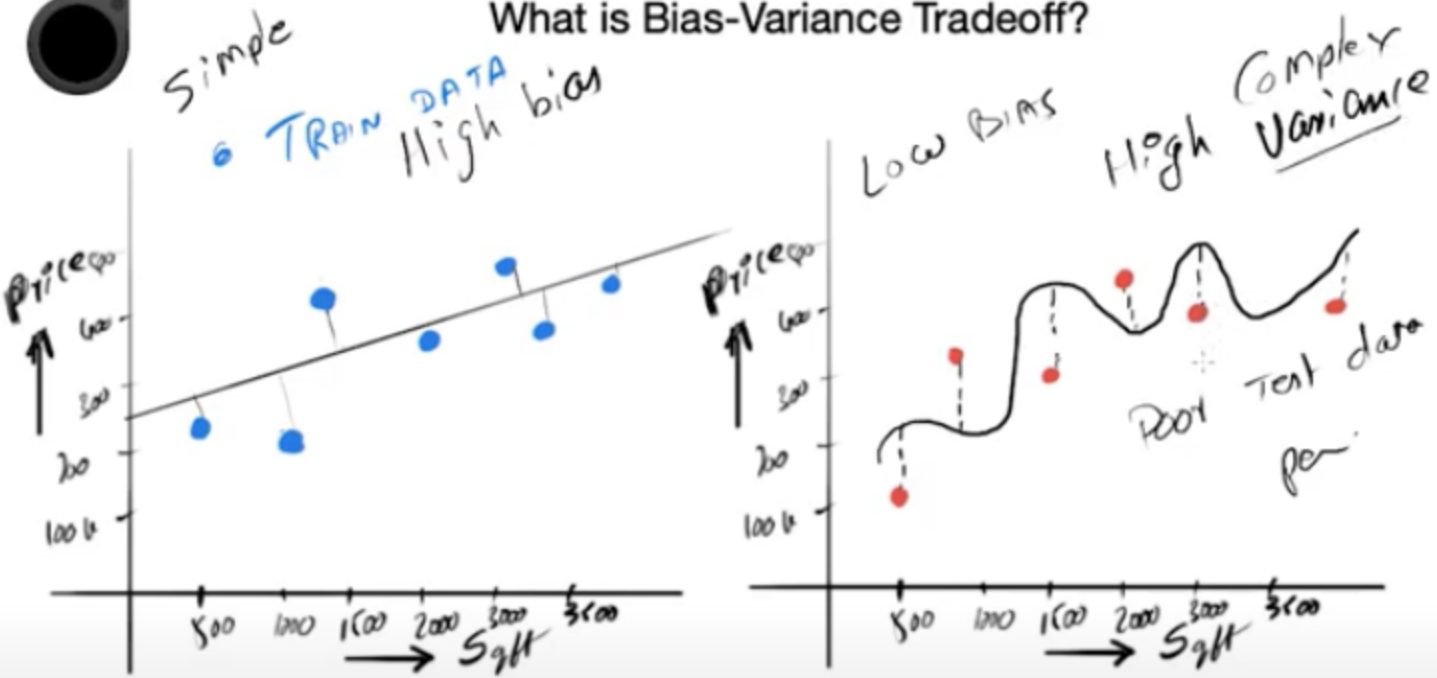


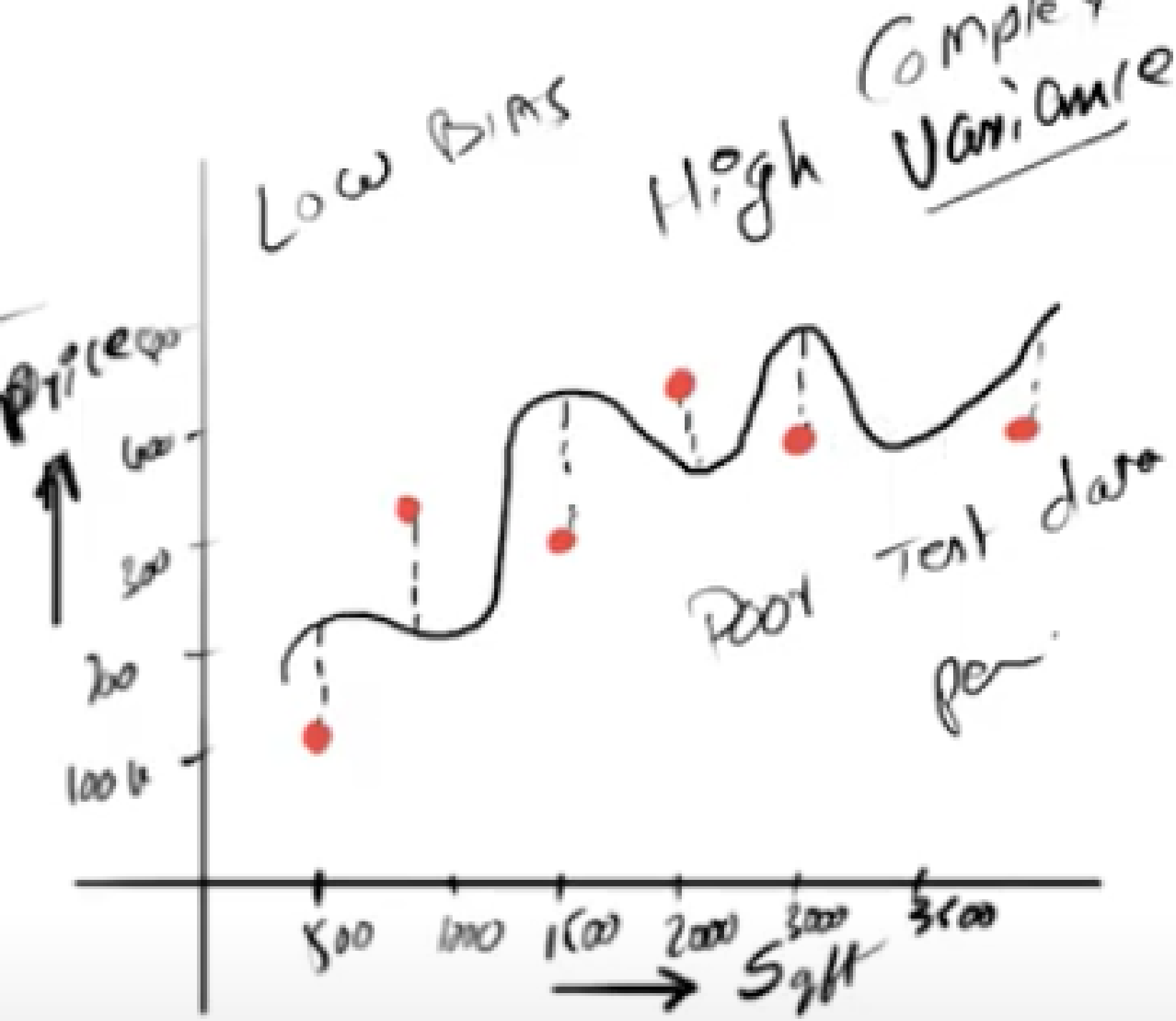
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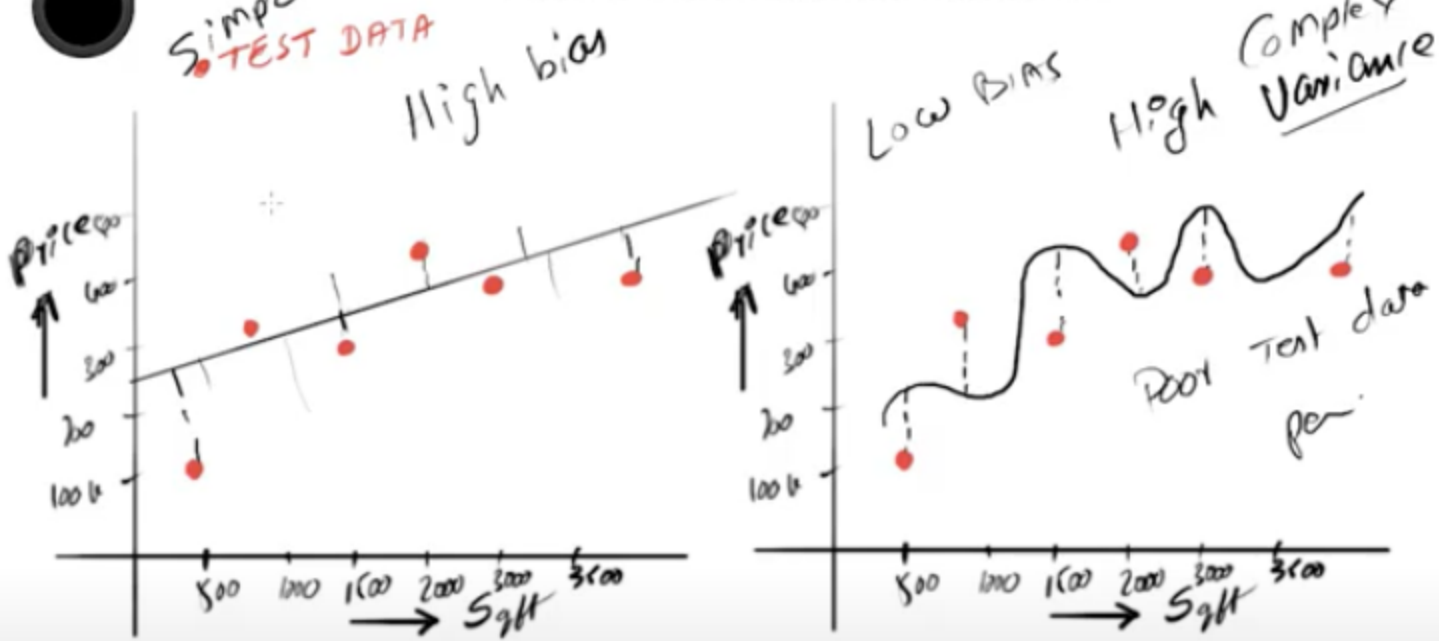

Low Bias
Price
600
300
200
100
500
1000
1500
2000
2500
3000
Sqft

What is Bias-Variance Tradeoff?

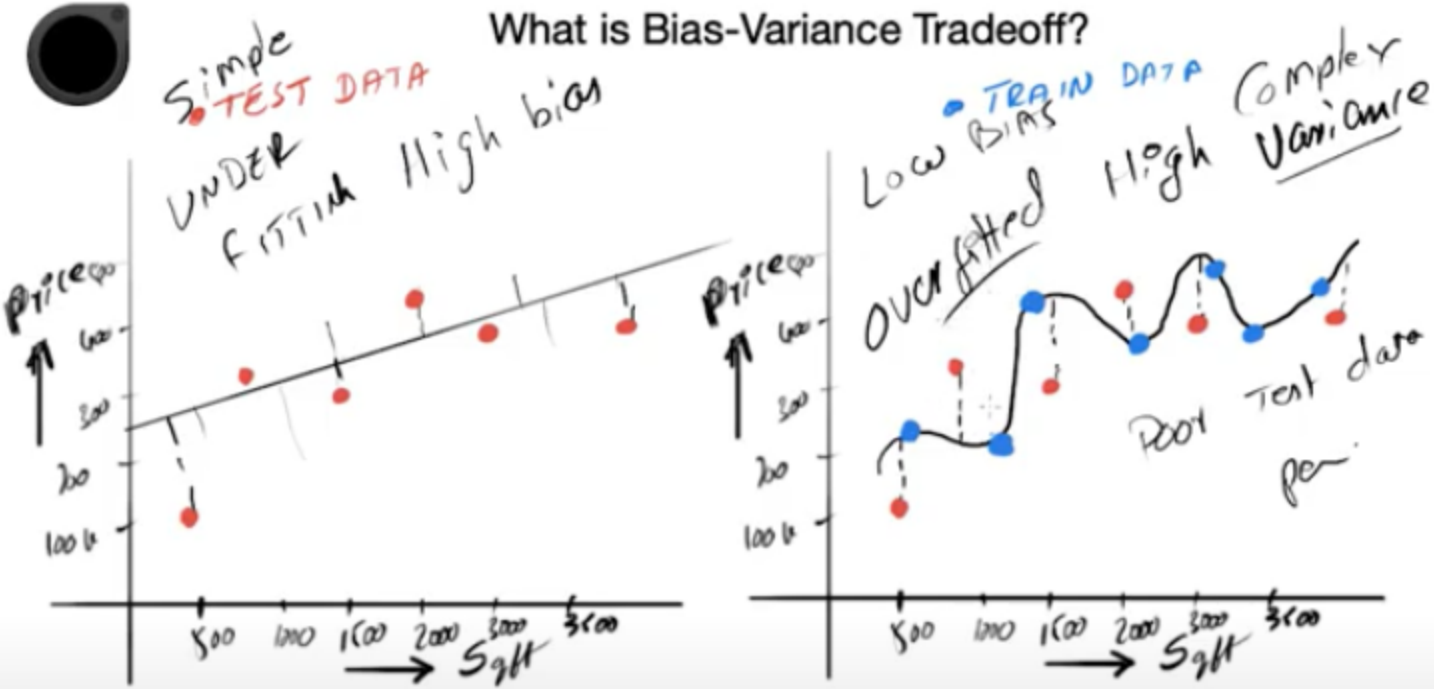



Low Bias
High Variance
Complex
Price
600
400
200
100
500
800
1000
1200
2000
3000
Sqft
Real Test data
per

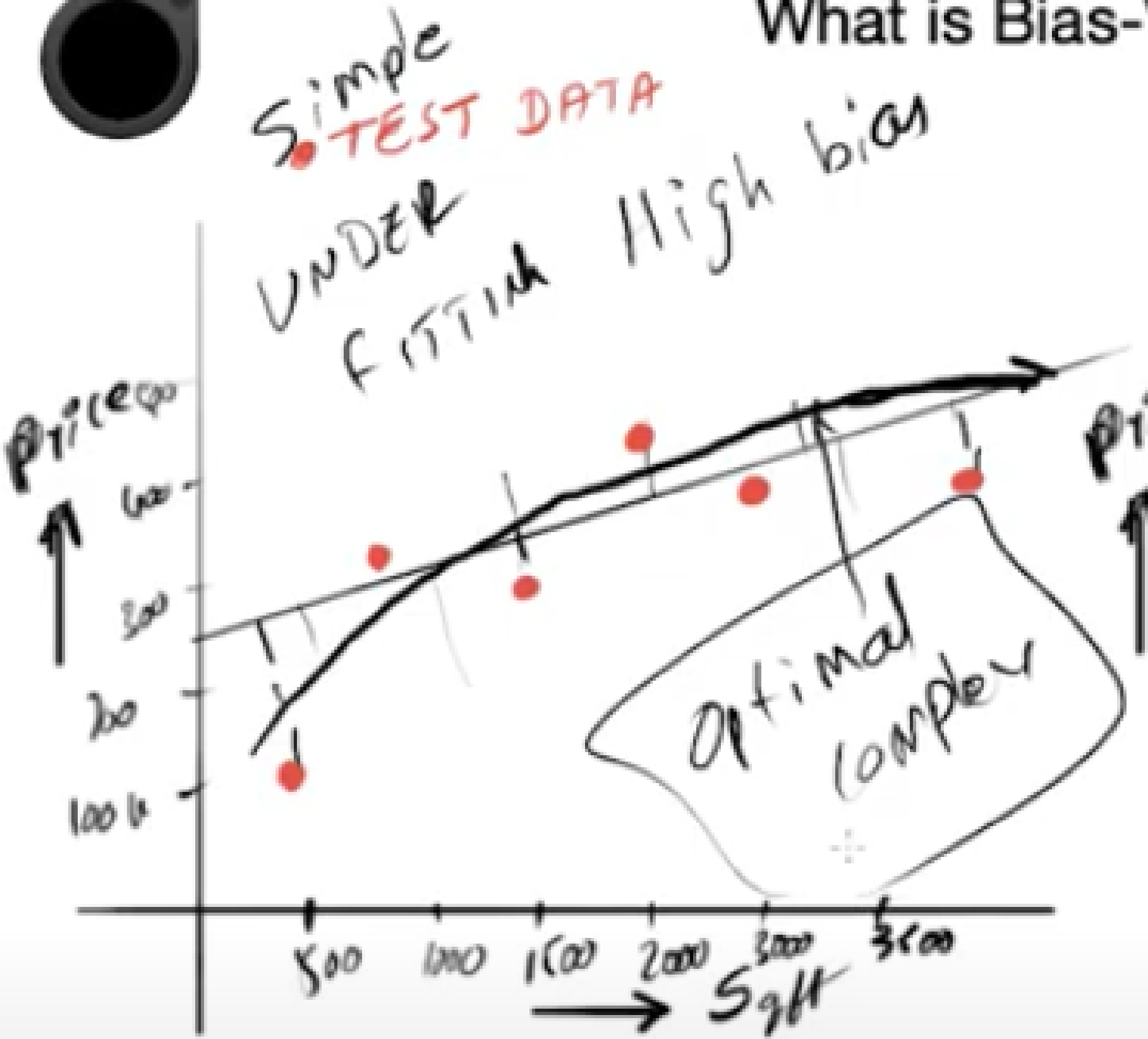
What is Bias-Variance Tradeoff?

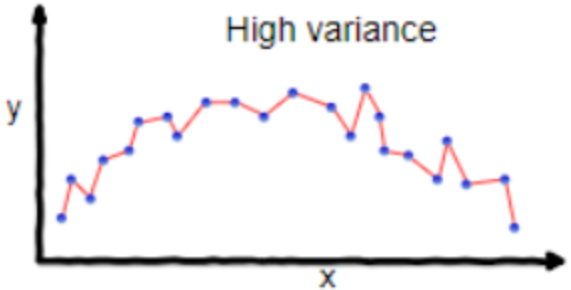


What is Bias-Variance Tradeoff?

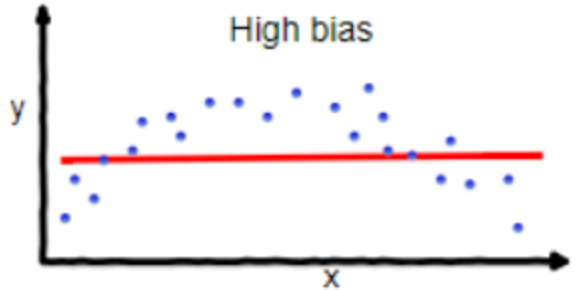


What is Bias-

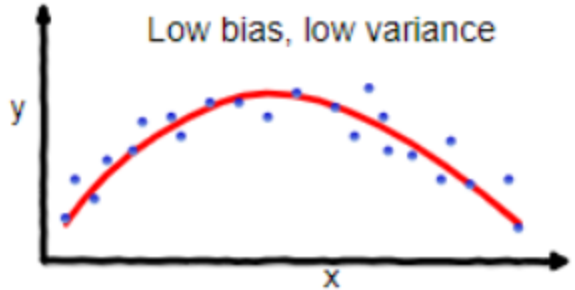

Simple
TEST DATA
UNDER
FITTING
High bias
Optimal
complexity



overfitting



underfitting



Good balance

What is bias?

Bias is the difference between the average prediction of our model and the correct value which we are trying to predict. Model with high bias pays very little attention to the training data and oversimplifies the model. It always leads to high error on training and test data.

What is variance?

Variance is the variability of model prediction for a given data point or a value which tells us spread of our data. Model with high variance pays a lot of attention to training data and does not generalize on the data which it hasn't seen before. As a result, such models perform very well on training data but has high error rates on test data.

Why is Bias Variance Tradeoff?

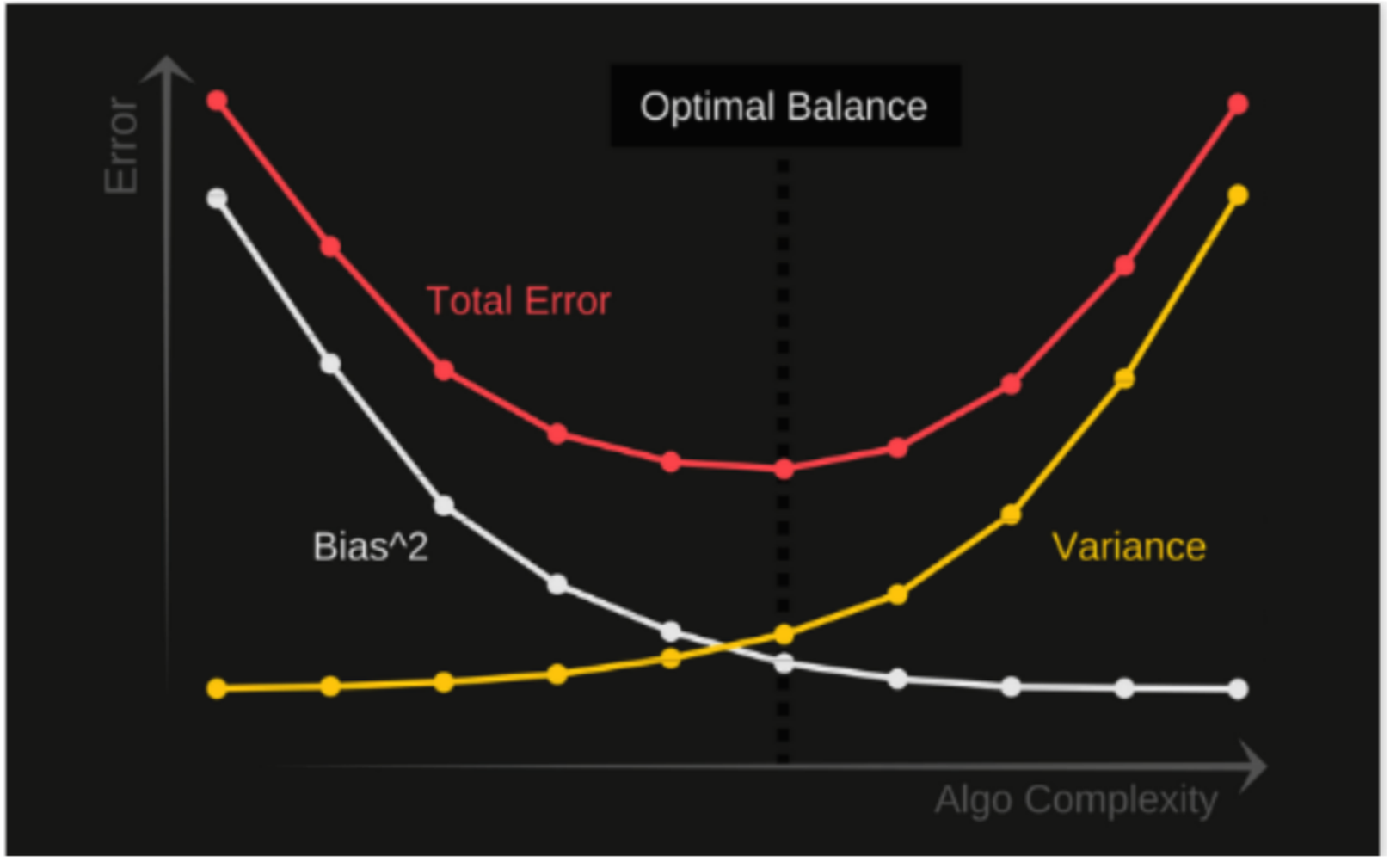
If our model is too simple and has very few parameters then it may have high bias and low variance. On the other hand if our model has large number of parameters then it's going to have high variance and low bias. So we need to find the right/good balance without overfitting and underfitting the data.

This tradeoff in complexity is why there is a tradeoff between bias and variance. An algorithm can't be more complex and less complex at the same time.

Total Error

To build a good model, we need to find a good balance between bias and variance such that it minimizes the total error.

Total Error = Bias^2 + Variance + Irreducible Error



An optimal balance of bias and variance would never overfit or underfit the model.

Therefore understanding bias and variance is critical for understanding the behavior of prediction models.